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Optimal Decision-Making Based on Prediction Sets

Risk-Optimal Conformal Prediction Derives the Policy and the Set from the Same Minimax Objective

By Tao Wang & Edgar Dobriban (University of Pennsylvania, Wharton Statistics and Data Science)

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A conformal prediction set guarantees that the true label falls inside it with probability at least $1 - \alpha$ —but says nothing about what to *do* once you have it. This paper derives the minimax-optimal downstream decision policy for a fixed prediction set, then works backward to the optimal set construction itself, unifying both into a single algorithm: **Risk-Optimal Conformal Prediction (ROCP)**. The result consistently cuts worst-case risk and critical mistake rates relative to standard max-min and best-response baselines, with the largest gains where out-of-set errors are most costly.

AT A GLANCE

Problem: Coverage guarantees say nothing about how to act optimally on a prediction set for a downstream decision.

Why it matters: Prediction sets are increasingly used to drive real decisions (triage, risk limits, control); an uncalibrated decision rule can waste or misuse the coverage guarantee.

Method: Minimax decision theory over distributions consistent with the coverage constraint, solved jointly for the optimal policy and the optimal prediction set.

Result: ROCP reduces worst-case risk and critical mistake rates versus max-min and best-response baselines, especially when out-of-set errors are costly.

Evidence: Comparisons against risk-averse max-min policies and naive best-response policies across decision tasks with varying error costs.

The Big Picture

Conformal prediction has become the standard way to wrap any black-box model with a distribution-free coverage guarantee: the resulting set $C(x)$ contains the true label with probability at least $1 - \alpha$. But in practice, sets are rarely the end product—they feed a downstream decision. A clinician acts on a set of possible diagnoses; a

control system acts on a set of possible next states. Until now, the choice of *how* to act on $C(x)$, and even how $C(x)$ should be shaped for that purpose, has been left to ad hoc heuristics disconnected from the coverage guarantee itself.

This paper treats the full pipeline—set construction and downstream decision—as one decision-theoretic problem, and solves for both pieces jointly under a minimax risk objective.

Why Existing Approaches Fall Short

Two heuristics dominate practice. **Best-response** policies act as if the point prediction were exact, ignoring the coverage guarantee outright—leaving decisions exposed exactly when the true label lands elsewhere in the set. **Max-min** policies plan for the single worst label anywhere in $C(x)$, regardless of how likely the coverage guarantee actually makes it—wasting performance hedging against possibilities the guarantee doesn’t privilege. Neither is derived from the same objective that produced the set in the first place, so neither is calibrated to the decision task.

Core Mechanism

The paper first fixes an arbitrary prediction set $C(x)$ and solves for the minimax-optimal policy: the one minimizing worst-case expected loss over all distributions still consistent with the set’s coverage guarantee. That optimal policy balances the worst-case loss for labels inside $C(x)$ against a penalty for the residual probability mass that could fall outside it. Holding this policy fixed, the authors then solve backward for the prediction set that minimizes the resulting robust risk subject to the coverage constraint—yielding a set shaped for the decision task, not merely for coverage.

Technical Formulation

For decision $a \in \mathcal{A}$ and loss $\ell(a, y)$, define the robust risk of policy π given set C as

$$R(\pi, C) = \sup_{Q: Q(y \in C(x)) \geq 1 - \alpha} \mathbb{E}_Q[\ell(\pi(C(x)), y)].$$

The minimax-optimal π^* balances $\sup_{y \in C(x)} \ell(\pi^*, y)$ against an α -weighted penalty for loss outside $C(x)$. The optimal set $C^*(x)$ then minimizes $R(\pi^*, C)$ subject to $P(y \in C(x)) \geq 1 - \alpha$. ROCP estimates the population quantities needed for π^* and C^* using any black-box probabilistic model, then calibrates a nonconformity threshold on held-out data so that $C^*(x)$ retains exact finite-sample marginal coverage under exchangeability.

Learning or Inference Procedure

1. Fit a black-box probabilistic model to estimate the population quantities entering the oracle minimax policy π^* .
2. Construct the candidate set $C^*(x)$ that minimizes the resulting robust risk under the model’s estimates.
3. Calibrate a nonconformity threshold on a held-out calibration set to enforce exact finite-sample marginal coverage.
4. At decision time, apply π^* to the calibrated set $C^*(x)$ to produce the action a .

What the Guarantee Says

Regardless of how well the black-box model estimates the oracle quantities, the calibration step guarantees $P(y \in C^*(x)) \geq 1 - \alpha$ in finite samples under exchangeability alone. The risk-optimality of π^* and C^* is with respect to the worst-case distribution consistent with that guarantee—so the decision quality improves as the underlying model improves, while the coverage guarantee never depends on the model being correct.

Experimental Findings

Across decision tasks with varying costs for out-of-set errors, ROCP reduces both worst-case risk and the rate of critical (high-cost) mistakes compared to the max-min baseline (over-conservative) and the best-response baseline (under-protected). The advantage widens as the cost of an out-of-set error grows, precisely the regime where a principled decision policy matters most.

Ablations and Interpretation

Cost sensitivity: When out-of-set costs are low, ROCP, max-min, and best-response converge in performance—the decision-theoretic refinement matters most when mistakes are expensive.

Set size: ROCP’s sets are not simply smaller or larger than standard conformal sets; they are reshaped toward the decision task, sometimes excluding low-cost-if-missed labels in favor of tighter coverage where mistakes are expensive.

Model quality: Decision quality scales with the black-box model’s accuracy, while the coverage guarantee remains exact even under model misspecification—the two guarantees are cleanly separated.

Reference

Tao Wang, Edgar Dobriban. **Optimal Decision-Making Based on Prediction Sets.** arXiv:2602.00989, February 2026. <https://arxiv.org/abs/2602.00989>